palsr: Projecting the Locations of Moving Actors for Spatial Models of Dyadic Interaction

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Abstract Many actors studied in the social sciences—armed groups, firms, diplomats, migrating populations—have no fixed geographic location, and the place at which two such actors interact is itself a meaningful outcome. Standard spatial tools assume actors sit at fixed points and therefore cannot be applied directly. The **palsr** package implements the Projected Actor Location (PALS) method of Kim et al. (2023), which uses exponential-smoothing weights over the spatiotemporal histories of a focal actor and its interaction partners to project where a mobile actor effectively "is" at any moment, and tunes those weights to predict the locations of future interactions. The package provides a validated event-data container, maximum-similarity parameter estimation by minimizing great-circle prediction error, projection of actor locations at arbitrary times, construction of the dyadic distance covariates used in downstream interaction models, and nonparametric bootstrap uncertainty with Rubin's Rules pooling. Performance-critical kernels are implemented in C++ via **Rcpp**. We illustrate the full workflow on a bundled dataset of armed-conflict events in Nigeria.

1 Introduction

A great deal of social-scientific inquiry is *dyadic*: the unit of analysis is a pair of actors, and a central question is whether, how, and where the two interact. In the study of geopolitical relations—alliances, conflict, cooperation, trade—one of the most robust predictors of interaction is the geographic distance between the actors involved (Gleditsch and Ward, 2001; Ward et al., 2007). When the actors are states with fixed capitals or centroids, this distance is trivial to compute and treat as an exogenous covariate.

A large and growing share of substantively important actors, however, are not fixed in space. Rebel groups, militias, terrorist organizations, peacekeeping deployments, and non-governmental organizations move through a theatre of operations over time, and the fine-grained, geocoded event data that now document them—such as the Armed Conflict Location and Event Data project (ACLED; Raleigh et al., 2010)—record *where each interaction happened* but assign the actors themselves no persistent coordinates. Treating these actors as fixed-location entities discards exactly the spatial dynamics that drive their interactions.

Existing spatial software for R does not fill this gap. General geospatial toolkits such as **sf** (Pebesma, 2018) and spatial point-process libraries such as **spatstat** (Baddeley and Turner, 2005) are powerful, but they operate on objects with given coordinates; they provide no way to *infer* a moving actor's effective location, let alone one tuned to predict future interactions. Conversely, the dyadic interaction models common in political science and economics take inter-actor distance as an input. Kim et al. (2023) introduced the Projected Actor Location (PALS) method to bridge this gap, and showed that PALS-based distances substantially improve the prediction of subnational conflict relative to naive location measures. That article, however, was accompanied only by replication scripts specific to a single application.

This paper introduces **palsr**, an R package that turns PALS into reusable, documented, and tested research software. The package lets applied researchers apply PALS to their own dyadic event data without re-deriving the estimator, construct the distance covariates needed by downstream interaction models, and propagate estimation uncertainty into those models through a principled bootstrap. After reviewing the method, we describe the package's design and walk through a complete analysis of armed-conflict events in Nigeria.

2 The PALS method

2.1 Projecting a location

PALS represents the location of a mobile actor at a prediction time as a recency-weighted average of the locations of events in which the actor, and its past interaction partners, were involved. Let i be a *focal* actor and t a prediction time. Write e for an event with location $g(e)$ (a longitude–latitude pair) and age $t - t(e)$ at time t. PALS uses only events that occurred strictly before t.

The projected location is a convex combination of two components. The *focal* component is a recency-weighted mean of the locations of i's own past events; the *alter* component is a recency-weighted mean of the locations of events involving the actors that i has interacted with (its *alters*). Formally,

$$g_i(t) = (1-\pi) \sum_{e \in E_i} W_i(e)\, g(e) \;+\; \pi \sum_{e \in E_k} W_k(e)\, g(e), \qquad \pi = \text{logistic}(\gamma + \eta\, v), \tag{1}$$

where E_i and E_k are the focal and alter event sets. The weights decay with event age in the manner of exponential smoothing (Gardner Jr., 2006),

$$W_i(e) \propto \left(\text{age}(e)^{\alpha} + \epsilon\right)^{-1}, \qquad W_k(e) \propto \left(\text{age}(e)^{\beta} + \epsilon\right)^{-1}, \tag{2}$$

normalized to sum to one within each component, with a small offset ϵ to keep contemporaneous events finite. The mixing weight π governs how much the projection leans on the alters rather than the focal actor's own history. It depends through an intercept γ and a slope η on a statistic v that compares the number of focal events n_i with the number of alter events n_k, $v = (n_i/n_k)^{1/\sqrt{n_k}}$, so that an actor with a sparse personal history but active partners can borrow location information from those partners.

The method therefore has four parameters: α (decay of the focal history), β (decay of the alter histories), and γ and η (the focal-versus- alter mixing weight). A reduced *one-parameter* model fixes $\pi = 0$, using only the focal history and estimating α alone; it is fast and, as we show below, frequently competitive.

2.2 Estimating the parameters

The parameters are chosen to maximize the spatial similarity between predicted and observed interaction locations. The predicted location of an interaction between two actors is the mean of their two projected locations, and prediction proceeds by "marching forward" through time: each observed event is predicted using only events strictly preceding it. The objective is the mean great-circle (Haversine) distance between predicted and observed locations,

$$Q(\theta) = \frac{1}{n} \sum_{e} d_{\mathrm{H}}\big(g(e),\, \hat{g}(e;\theta)\big), \tag{3}$$

minimized over the parameter vector θ with `stats::optim` (Brent's method for the one-parameter model, Nelder–Mead for the full model). Because the projection is recomputed for every event at every candidate θ, estimation re-projects every actor across a grid of times many times over; the inner projection and distance kernels are implemented in C++ for speed (see below).

3 The palsr package

3.1 Event data

The central data object is a `pal_events` table, a validated `data.frame` subclass with one row per dyadic event and canonical columns `actor1`, `actor2`, `time` (a `Date`), `lon`, and `lat`. The

Function	Purpose
`pal_events()`	Build and validate a dyadic event table
`pals_params()`	Specify a parameter set directly
`estimate_pals()`	Estimate parameters by Haversine-error minimization
`project_pal()`, `project_pals()`	Project actor locations at given times
`predict_event_locations()`	Predict dyadic interaction locations
`pal_distance()`	Dyadic distance between projected locations
`bootstrap_pals()`	Nonparametric bootstrap uncertainty
`pool_rubin()`	Pool replicate estimates with Rubin's Rules
`simulate_conflict_events()`	Simulate mobile-actor interaction data

Table 1: The main user-facing functions of **palsr**.

constructor `pal_events()` maps arbitrary column names onto this schema, coerces types, checks coordinate ranges, drops incomplete or self-dyadic rows, and sorts by time:

```
library(palsr)
raw <- data.frame(from = c("A", "A", "B"), to = c("B", "C", "C"),
                  when = as.Date(c("2001-01-01", "2001-06-01", "2002-01-01")),
                  x = c(7.1, 8.0, 7.5), y = c(9.0, 9.4, 10.1))
ev <- pal_events(raw, actor1 = "from", actor2 = "to",
                 time = "when", lon = "x", lat = "y")
```

3.2 Main functions

The package exposes the PALS workflow through a small set of functions, summarized in Table 1.

A model is fit with `estimate_pals()`, which returns an object of class `pals_fit` with `print`, `summary`, `coef`, and `predict` methods. Given a fit (or a hand-specified `pals_params` object), `project_pals()` returns the projected location of each actor at one or more times, `predict_event_locations()` predicts interaction locations and scores them against observed coordinates, and `pal_distance()` constructs the dyadic distance covariate—optionally log-transformed—that enters downstream models of who interacts with whom.

3.3 Implementation

The projection kernel and the great-circle distance are implemented in C++ and exposed through **Rcpp**. A batched kernel evaluates the projection for all events of an estimation problem in a single compiled call, so that each evaluation of the objective during optimization avoids the overhead of repeated R-level iteration. The pure-R reference implementation of the kernel is retained and is checked against the compiled version in the package's test suite, which uses **testthat**.

3.4 Bundled data

So that the full workflow can be learned and reproduced without external data, the package bundles two datasets. `nigeria_acled` contains 1,549 dyadic conflict events among 37 actors in Nigeria between 1997 and 2016, drawn from the public replication archive for Kim et al. (2023); the underlying event records come from ACLED. `nigeria_sim` is a deterministic, seeded simulation produced by `simulate_conflict_events()`, which generates events among actors that drift slowly through space and interact preferentially with nearby partners.

4 Illustration

We illustrate **palsr** with the bundled nigeria_acled data. We first fit the one-parameter model, which estimates only the focal decay α:

```
data(nigeria_acled)
fit1 <- estimate_pals(nigeria_acled, model = "one")
coef(fit1)
#>    alpha
#> 1.240806
```

The fitted $\alpha \approx 1.24$ implies steeply decaying weights, so a projected location is dominated by an actor's most recent events. At the optimum the mean prediction error is about 147 km over the 1,540 events with usable history. The full four-parameter model adds the alter component:

```
fit4 <- estimate_pals(nigeria_acled, model = "four")
coef(fit4)
#>       alpha        beta       gamma         eta
#>  1.24058   0.01258   -11.32243    -9.70853
```

Here the large negative γ drives the mixing weight π to essentially zero, so the four-parameter fit reproduces the one-parameter fit almost exactly (an objective of 147 km in both cases). This recovers a substantive finding of Kim et al. (2023): in this application an actor's own recent history is far more informative about where it will next be engaged than the histories of its partners, and the parsimonious one-parameter model loses little.

Because the projection is recomputed as time advances, each actor traces a *trajectory* through space. Figure 1 shows the projected paths of four actors at yearly intervals, drawn over the cloud of observed events:

```
actors <- c("Boko Haram", "Fulani Ethnic Militia (Nigeria)",
            "Ijaw Ethnic Militia (Nigeria)")
dates  <- as.Date(sprintf("%d-01-01", 2005:2016))
traj   <- project_pals(nigeria_acled, actors = actors,
                       predict_time = dates, params = fit1)
```

The projected locations feed directly into models of dyadic interaction. The function pal_distance() returns the great-circle distance between two actors' projected locations at a given time, the key spatial covariate of Kim et al. (2023):

```
dyads <- data.frame(actor1 = "Boko Haram",
                    actor2 = "Fulani Ethnic Militia (Nigeria)",
                    time   = as.Date("2014-06-01"))
pal_distance(nigeria_acled, dyads, fit1, transform = "log")
#>     actor1                     actor2      time pal_distance pal_log_distance
#>  Boko Haram Fulani Ethnic Militia (Nig) 2014-06-01       445.6           6.099
```

4.1 Quantifying uncertainty

Parameter and projection uncertainty are quantified by a nonparametric bootstrap. bootstrap_pals() resamples events with replacement, re-estimates the model on each replicate, and returns bootstrap standard errors and percentile intervals:

```
bt <- bootstrap_pals(nigeria_acled, R = 20, model = "one", seed = 1)
summary(bt)
#>   term estimate boot_mean boot_se lower upper
#>  alpha    1.241     1.240   0.100 1.075 1.451
```

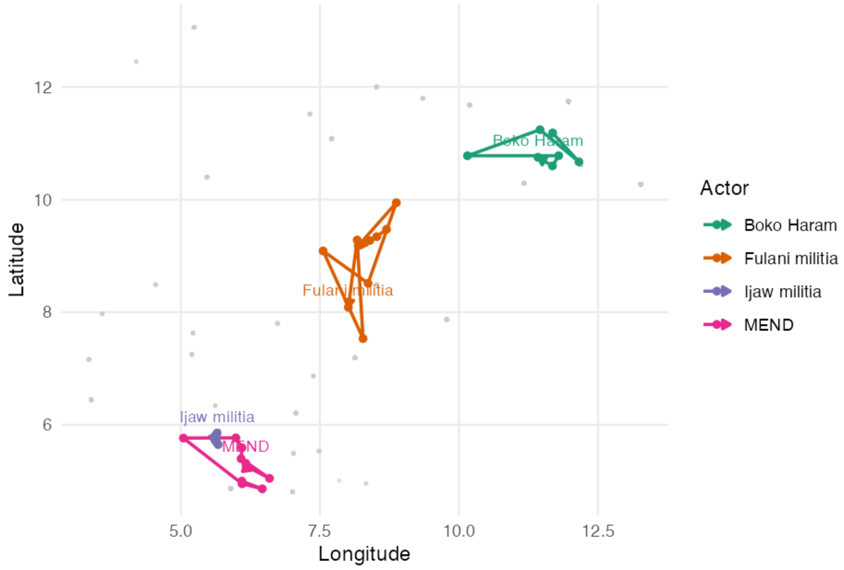

Figure 1: PALS-projected trajectories of four actors in the `nigeria_acled` data, fit with the one-parameter model and projected at yearly intervals from 2005 to 2016. Grey points are the observed dyadic events; coloured paths trace each actor's projected location over time, with arrows pointing forward in time. The projections separate the actors into distinct, drifting regions of operation.

The 95% percentile interval for α is roughly $[1.08, 1.45]$, comfortably away from zero. When a downstream estimand—for example a regression coefficient in a model of conflict that uses PAL distances—is computed on each replicate, the replicates can be treated as multiple imputations and combined with Rubin's Rules (Rubin, 1987) using `pool_rubin()`, which propagates both within- and between-replicate uncertainty and optionally reports Barnard–Rubin degrees of freedom.

5 Summary

palsr provides a complete, tested implementation of the Projected Actor Location method for analysts who study interactions among geographically mobile actors. It turns a method that previously existed only as application-specific replication code into general-purpose software: validated data structures, maximum-similarity estimation with compiled kernels, projection and dyadic-distance construction, and bootstrap-based uncertainty quantification, together with bundled data and a vignette so the workflow can be reproduced end to end. The package is available from CRAN at `https://CRAN.R-project.org/package=palsr` and is developed openly at `https://github.com/bdesmarais/palsr`; its replication archive and the underlying method are documented in Kim et al. (2023).

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